

Omkar Chaskar

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Introduction

Computer Engineering graduate with hands-on experience in Generative AI, Prompt Engineering, LLMs, RAG pipelines, Deep Learning, and AI automation workflows. Skilled in building intelligent AI systems using LangChain, FAISS, Transformers, Python, Spring Boot, and REST APIs. Experienced in data annotation, vector databases, automation pipelines, and scalable backend systems with strong problem-solving and analytical skills.

Experience

Ajinkya Surface Coating

Java Backend Developer (Freelance)

Pune

Aug 2025 – Nov 2025

- Developed backend services using Java 17, Spring Boot, REST APIs, Hibernate, and MySQL.
- Implemented JWT authentication, layered architecture, pagination, validation, and exception handling.
- Optimized SQL queries and improved API performance for inventory and workflow systems.
- Worked with enterprise backend development practices and scalable application design.

Projects

Enterprise RAG System using LangChain & FAISS

Python, LangChain, FAISS, Transformers

- Built an end-to-end Retrieval-Augmented Generation (RAG) system using LangChain, FAISS, and transformer-based embeddings.
- Implemented document chunking, semantic retrieval, vector storage, prompt engineering, and LLM integration.
- Designed scalable intelligent question-answering pipelines for enterprise documents and unstructured datasets.
- Improved retrieval relevance by optimizing chunking strategy and embedding workflows for context-aware responses.

AI Email Code Review Automation using n8n

n8n, Gmail API, OpenAI API, LLMs

- Developed AI-powered automation workflow using n8n, Gmail API, and LLM integration for automated code analysis.
- Built workflow that reads incoming emails containing source code, summarizes logic, and auto-generates explanatory comments.
- Reduced manual code review time by automatically generating structured code summaries for developers.
- Integrated prompt engineering techniques and AI agents to improve readability and automation accuracy.

Melanoma Skin Disease Detection System

CNN, TensorFlow, Python, Deep Learning

- Developed deep learning-based melanoma detection system using CNN architecture and the ISIC Skin Cancer Dataset.
- Applied preprocessing, augmentation, feature extraction, and hyperparameter tuning for improved prediction accuracy.
- Improved model accuracy from 84% to 93% using data augmentation, dropout regularization, and optimized CNN layers.
- Implemented disease prediction and analysis workflows for intelligent skin disease classification.

Technical Skills

Programming: Java, Java, JavaScript, SQL

Generative AI: LLMs, Prompt Engineering, RAG, LangChain, Vector Databases, Semantic Search

Deep Learning: ANN, CNN, RNN, TensorFlow, Neural Networks

AI Automation: n8n, Gmail API, OpenAI API Workflow Automation

Data Science: Pandas, NumPy, Scikit-learn, EDA, Feature Engineering, Data Visualization

Data Annotation: HVAT, Data Labeling, Dataset Preparation, Annotation Workflows

Backend Development: Spring Boot, REST APIs, Hibernate, JWT Authentication, MySQL

Database & ETL: MySQL, JDBC, DBMS, ETL Pipelines, Data Cleaning

Core Concepts: DSA, OOP, System Design, OS, Computer Networks, HTTP/HTTPS

Tools: Git, GitHub, Postman, VS Code, Jupyter Notebook, Linux

Education

Sinhgad College of Engineering
B.E. Information Technology

2025
CGPA: 8.0

Certifications

- Introduction to Large Language Models – IBM
- Anthropic Claude Network Path